Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/02/2019
PI Name:	Goree, John Ph.D.		
Project Title:	Nonlinear Wave Experiments in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	06/14/2017	**End Date:**	06/13/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Change in start and end dates to 6/14/2017-6/13/2021 (from 4/15/2017-4/15/2021) per I. Hahn/JPL and PI (Ed., 1/19/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liu, Bin Ph.D. (University of Iowa)		
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Performance Goal Text:			

Task Description:

A flight experiment will be performed and analyzed in a study of nonlinear synchronization of waves. Synchronization is a cooperative phenomenon for oscillators or waves. A signature of synchronization is a frequency that shifts when an external modulation is applied. Compared to point-like oscillators, waves are more complicated due to their spatio-temporal behavior and dispersion properties. An understanding of synchronization is lacking for waves of all kinds, including those in dusty plasmas.
The experiment will use dusty plasmas, also called "complex plasmas." These are mixtures of small particles of solid matter, electrons, ions, and neutral gas. The small solid particles obtain large electric charges, so that they exhibit collective behavior. They are electrically confined, so that they do not contact any boundaries.

Microgravity enables this experiment by suppressing sedimentation, thereby allowing the waves to propagate distances of many wavelengths, with relatively uniform background conditions. This experiment is based on previous ground-based experiments, where waves propagated only a few wavelengths.

Our flight experiment will be performed using PK-4, which is a facility in the Columbus Module of the International Space Station (ISS), built by the European Space Agency.

SCIENTIFIC/TECHNICAL MERITS

The main scientific topic is the synchronization of a dust acoustic wave (DAW). A DAW is a low-frequency compressional wave, which involves charge-separation from electrons and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Synchronization experiments are performed by applying electrical waveforms to an electrode to modulate the dusty plasma sinusoidally. The diagnostic for detecting the wave motion is video imaging using particle observation cameras. Image data will be analyzed to prepare space-time diagrams, which provide a spatial and temporal description of the wave's amplitude as it propagates. These space-time diagrams are the foundation for further analysis methods.

In support of our experiment, a theoretical effort will be undertaken at Iowa. The goal is to contribute to the development of a theory for synchronization of propagating waves, to compare to the experiment. The theoretical effort at Iowa will focus on numerical simulations using a hybrid model. Molecular dynamics simulation will also be performed. The main theoretical effort, which will center on analytic theory, will be done by theorist elsewhere, and we will collaborate with them.

The feasibility of our flight experiment is based on our previous ground-based experiments, which confirm that synchronization of the DAW occurs. These ground-based experiments also revealed unexplained phenomena, including a nonharmonic synchronization state, which we will investigate with our flight data.

In the unlikely event that our main experiment with waves does not yield useful results, we have a plan for a backup experiment to investigate mobility-limited motion of particles having a different size.

BROADER IMPACTS

This research has an overlap with nonlinear dynamics as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, a website, and Twitter. A K-12 outreach presentation will be made at an elementary school.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

During this period, our group carried out flight experiments and ground-based research, investigating nonlinear waves in three-dimensional dusty plasmas under microgravity conditions. Our first ISS flight experiment was successfully completed in PK-4 Campaign 5. It investigated the synchronization of nonlinear waves in three-dimensional dusty plasmas. In preparation for a follow-up ISS flight experiment on the synchronization, a parabolic flight experiment was also performed, using the PK-4 parabolic flight model at University of Giessen, Germany. Other research activities included flight experiment planning, analysis method development, and theoretical support. For our research group, our research activities have led to six peer-reviewed journal papers, eight contributed talks, and one Ph D thesis.
Our research focused on the fundamental physics of dusty plasma. Small particles of solid matter, also called dust, are immersed in a mixture of electrons, ions, and neutral gas, making a dusty plasma. These small particles obtain large electric charges, so that they exhibit collective behavior like waves and viscous transport. In this project, we investigated the fundamental nature of a nonlinear wave called dust acoustic wave (DAW).

NONLINEAR DUST ACOUSTIC WAVES

A DAW is a low-frequency compressional wave, which involves charge-separation among the dust, electrons, and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Two nonlinear effects were studied: nonlinear waveforms and nonlinear synchronization. A signature of synchronization is a frequency shift due to an external modulation.

OUR FIRST ISS FLIGHT EXPERIMENT

Iowa’s first ISS flight experiment was successfully completed in PK-4 Science Campaign 5, November 2018. This flight experiment investigated nonlinear waves in three-dimensional dusty plasmas under microgravity conditions. It successfully accomplished its goals:

• finding parameter regime for the excitation of steady nonlinear dust acoustic waves under microgravity conditions;

• testing modulation methods for wave synchronization.

Co-Investigator Bin Liu participated in the execution of this experiment in Toulouse France, together with PK-4 Core-Team scientists. He also observed all other PK-4 flight experiments performed in the Campaign 5.

To prepare for this flight experiment, we worked closely with PK-4 Core Team scientists and prepared our flight experiment. At Iowa, we designed our experiment procedure, configuration, and parameters regime, yielding a flow chart. PK-4 Core Team generated an experiment script based on our flow chart. Co-I Bin Liu travelled to the DLR

Task Progress:	(German Space Agency), and tested the script using the PK-4 ground model at the DLR. This test resulted in the final version of our flight experiment script, which was executed in the PK-4 Science Campaign 5. EUROPEAN SPACE AGENCY (ESA) PK-4 PARABOLIC FLIGHT CAMPAIGN As a preparation for future experiments on the ISS, we participated in the 12th PK-4 Parabolic Flight Campaign in May 2019, investigating complex (dusty) plasmas under microgravity. This campaign used the PK-4 parabolic flight model at University of Giessen. Thirty-one parabolas (one day) were dedicated for two Iowa experiments: steady dust acoustic waves and pulse shear motion. The flight experiment scripts were executed successfully, and all planned tests were accomplished. To prepare for this parabolic flight campaign, Co-I Bin Liu traveled to University of Giessen and worked with scientists at Giessen, yielding flow charts and experiment scripts for Iowa experiments. CNOIDAL WAVEFORM OF NONLINEAR DUST ACOUSTIC WAVES To develop the analysis method to be used with our ISS flight experiment, we analyzed previous breadboard experiments, involving nonlinear waves. We characterized the time variation of the waves, as observed from the ground-based experiments. This analysis effort, in collaboration with the theorists at NRL (Naval Research Laboratory), resulted in a paper, which was recently published by Physics of Plasmas. This paper was chosen by the Editor to be promoted as an Editor's Pick. We found that the experimentally measured waveform of nonlinear dust acoustic waves can be accurately described by a Cnoidal function. Fitting the experimental waveforms to the Cnoidal function also provides a measure of the wave's nonlinearity. The underlying theory that predicts the Cnoidal waveform as an exact solution of a Korteweg-de Vries model equation takes account of the streaming ions that are responsible for the spontaneous excitation of the dust acoustic waves. PK-4 ISS FLIGHT DATA Research activities on-site at Iowa also included a survey of the flight data from previous missions. During this report period, we completed an analysis of the flight data from Science Campaign #2, yielding a paper published in a peer-reviewed journal. This is our second paper that uses PK-4 flight data. During the past year, the Iowa group received the flight data from Science Campaigns 4 and 5. We also successfully obtained the PK-4 ground data including the quick-look data for previous missions. We have so far surveyed the data for most missions. To expedite our data analysis, we also hired one undergraduate, who was funded by National Science Foundation (NSF). FLIGHT EXPERIMENT PLANNING Our planning involved a collaboration with other institutions or agencies. Significant collaborators included PK-4 Core Team members in Germany and Russia, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, CADMOS (Microgravity and Spatial Operations Development Support Center) at the CNES (France National Center for Space Studies), and the collaborators specific to Iowa projects. Both PI John Goree and Co-I Bin Liu interacted with these collaborators frequently via email and occasionally by phone, and we also met them at various meetings and conferences. One significant meeting was the recent PK-4 workshop at the DLR (German Space Agency) in February 2019. Co-I Bin Liu attended this meeting, and he gave a talk regarding the progress of Iowa PK-4 projects. During this workshop, Liu also discussed our plan for the next flight experiment with Core Team scientists, on the topic of the shear motion induced by a manipulation laser. BROADER IMPACTS Our research results have been disseminated in journals and at conferences. During this report period, our research group has published six papers in peer-reviewed journals and given seven contributed talks in various conferences.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
Abstracts for Journals and Proceedings	Kananovich A, Goree J. "Particulate imaging diagnostics in a gas-discharge plasma." Presented at the 71st Annual Gaseous Electronics Conference, Portland, OR, November 5-9, 2018. 71st Annual Gaseous Electronics Conference, Portland, November 5 - 9, 2018. , Nov-2018
Abstracts for Journals and Proceedings	Kananovich A, Goree J. "Experimental measurement of shock thickness in a strongly coupled dusty plasma." Presented at the 71st Annual Gaseous Electronics Conference, Portland, OR, November 5-9, 2018. 71st Annual Gaseous Electronics Conference, Portland, November 5 -9, 2018. , Nov-2018
Abstracts for Journals and Proceedings	Liu B, Goree J. "Shear deformations in dusty plasma." Presented at the 15th Workshop on the Physics of Dusty Plasmas, Baltimore, MD, May 29- June 1, 2018. Abstracts. 15th Workshop on the Physics of Dusty Plasmas, Baltimore, May 29-June 1, 2018. , Jun-2018
Abstracts for Journals and Proceedings	Wei Z, Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Diffusive motion in a three-dimensional cluster in PK-4." Presented at the 15th Workshop on the Physics of Dusty Plasmas, Baltimore, MD, May 29-June 1, 2018. Abstracts. 15th Workshop on the Physics of Dusty Plasmas, Baltimore, May 29-June 1, 2018. , Jun-2018
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Articles in Peer-reviewed Journals	Liu B, Goree J, Flanagan T, Sen A, Tiwari S, Ganguli G, Crabtree C. "Experimental observation of cnoidal waveform of nonlinear dust acoustic waves." Physics of Plasmas. 2018 Nov;25(11):113701. https://doi.org/10.1063/1.5046402 , Nov-2018
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Dissertations and Theses	Wong CS. "Statistical physics principles tested using dusty plasma and aerosol experiments." Dissertation, The University of Iowa, Iowa City, August 2018. , Aug-2018